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glucose into two molecules of pyruvic acid. Acetyl CoA formed from pyruvic acid, acts as a link between glycolysis and Krebs cycle. Krebs cycle takes place in matrix of mitochondria and also called as citric acid cycle in which CO_2 and H_2O were produced. Hydrogen removed from the substrates is received by coenzymes which get reduced. They are again oxidised by removal of hydrogen. This hydrogen splits into protons and electrons. The electrons transferred through various electron transport carriers present in inner membrane of mitochondria is used for the synthesis of ATP with the help of ATP synthase. This process is called **oxidative phosphorylation**.

Anaerobic respiration involves incomplete breaking down of the substrate glucose into ethyl alcohol or lactic acid. In aerobic respiration 36 ATP molecules are produced in plant mitochondria but in animals 38 ATP molecules are produced per glucose molecule. During anaerobic respiration only 2 ATP molecules are produced, therefore anaerobic respiration is less efficient than aerobic respiration. The respiratory quotient (RQ) is the ratio of carbon dioxide production to oxygen consumption and reflects the relative contributions of fat, carbohydrate, and protein to the oxidation. Pentose phosphate pathway is an alternative pathway to glycolysis and TCA cycle for oxidation of glucose. It occurs in cytoplasm of both prokaryotes and eukaryotes.

Evaluation

CR2HJX

1. The number of ATP molecules formed by complete oxidation of one molecule of pyruvic acid is

   a. 12 b. 13 c. 14 d. 15

2. During oxidation of two molecules of cytosolic NADH + H^+, number of ATP molecules produced in plants are

   a. 3 b. 4 c. 6 d. 8

3. The compound which links glycolysis and Krebs cycle is

   a. succinic acid b. pyruvic acid
   c. acetyl CoA d. citric acid

4. Assertion (A): Oxidative phosphorylation takes place during the electron transport chain in mitochondria.
   Reason (R): Succinyl CoA is phosphorylated into succinic acid by substrate phosphorylation.

   a. A and R is correct. R is correct explanation of A
   b. A and R is correct but R is not the correct explanation of A
   c. A is correct but R is wrong
   d. A and R is wrong.

5. Which of the following reaction is not involved in Krebs cycle.

   a. Shifting of phosphate from 3C to 2C
   b. Splitting of Fructose 1,6 bisphosphate of into two molecules 3C compounds.
   c. Dephosphorylation from the substrates
   d. All of these

6. What are enzymes involved in phosphorylation and dephosphorylation reactions in EMP pathway?
7. Respiratory quotient is zero in succulent plants. Why?
8. Explain the reactions taking place in mitochondrial inner membrane.
9. What is the name of alternate way of glucose breakdown? Explain the process involved in it?
10. How will you calculate net products of one sucrose molecule upon complete oxidation during aerobic respiration as per recent view?